

Analyzing the Correlation Between Ski Pole Design and Buckling

Introduction:

When initially brainstorming project ideas, our group wanted to explore a topic that we could relate to.

Given Mari's experience with cross-country skiing, we decided to explore ski poles, looking specifically at the material behavior of poles given loading conditions. Upon further research, we found many types of ski poles that are used for different purposes/preferences (Figure 1). As a result, we were curious about the designs of different ski poles and why they had varying levels of bending throughout their frames. Our initial assumption was that this bending may cause forces to be propagated

differently, each pole designed specifically for unique

loading cases. Thus, we strove to answer the question: How does ski pole design influence force propagation under load conditions? Specifically, we looked at how the critical buckling force differed in two different ski pole designs to see which design was better suited to supporting a higher buckling force. By understanding this force, we could then more closely look at the use cases in real life of different poles and why specific individuals and their use cases would benefit from a specific design.



Figure 1: Ski pole design variation

Design:

The most important design choice we made in this project was purposefully choosing to design two different ski poles: one that was completely straight and another in which the frame was significantly bent (Figure 2). These poles were modeled off of the first and third poles seen in Figure 1, respectively. After designing and creating CAD models of these two ski poles, they were essentially identical to a life-sized model of these poles.

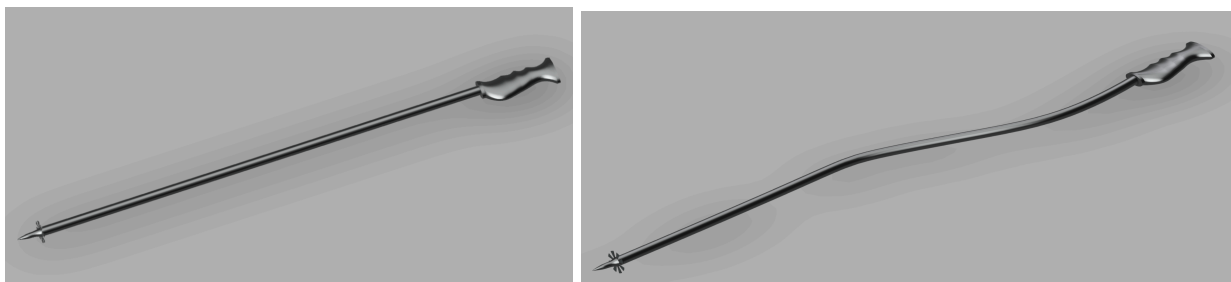


Figure 2: Straight pole life-sized scale CAD (left); Bent pole life-sized scale CAD (right)

For our testing, we fully designed, calculated, and ran finite element analyses (FEA) on both full-sized and scaled-down models. However, since we had a manufacturing constraint, we were unable to fabricate the life-sized model. Thus, when we created the scaled-down model, we purposefully kept the design essentially the same as our full-scale model to ensure our results were as close to real life as possible, besides removing the pointed tip and flange to reduce extra variables to account for in testing. When choosing our scale for the fabrication, we wanted to make it as large as possible given the constraint of the 3D printing bed, which came out to be roughly 22 cm long. As a result, our fabricated models were scaled down to 0.1833 times the original design. For the remainder of the report, we have included the hand calculations and FEA for both sized models and only the physical testing results for the scaled-down model.



Figure 3: 3D print of physical prototypes (left); Initial physical prototypes (right)

Analytical Results:

For our analytical results, we could only calculate the critical buckling load for the straight pole. This was mainly due to the fact that given the equations we learned in class (and what we were able to find online) there was no feasible way to simply calculate the critical buckling load for the bent pole. Thus, this section highlights the analytical results for our life-sized model and scaled prototype.

For our life-sized ski pole analysis, we assumed that the radius of the pole was constant throughout the length of the pole, the bottom of the rod was fixed to the ground, and the top of the rod was free (for ease of calculation to FEA comparison). Additionally, we neglected the handle and the sharp tip of the ski pole, replacing it with an extension of the rod itself. This resulted in a rod that was 1.2 m long, 2 cm in diameter, 2.5 mm wall thickness, and made out of Aluminum 6061.

For our scaled-sized ski pole analysis, we assumed that the radius of the pole was constant throughout the length of the pole, the bottom of the rod was pinned to the ground (to reflect testing conditions), and the top of the rod was free (to reflect testing conditions). Given this boundary condition, we settled on the K-value to be 1.5. Our rationale behind this stemmed from a pinned-pinned system having a K-value of 1 and a fixed-free system having a K-value of 2. Thus, we met in the middle with a pinned-free K-value of 1.5 to reflect our physical testing boundary conditions. Again, we neglected the handle and the sharp tip of the ski pole, replacing

it with an extension of the rod itself. This resulted in a rod that was 22 cm long, 3.8 mm in diameter, 0.475 mm wall thickness, and made out of Polylactic acid (PLA)—to reflect testing conditions.

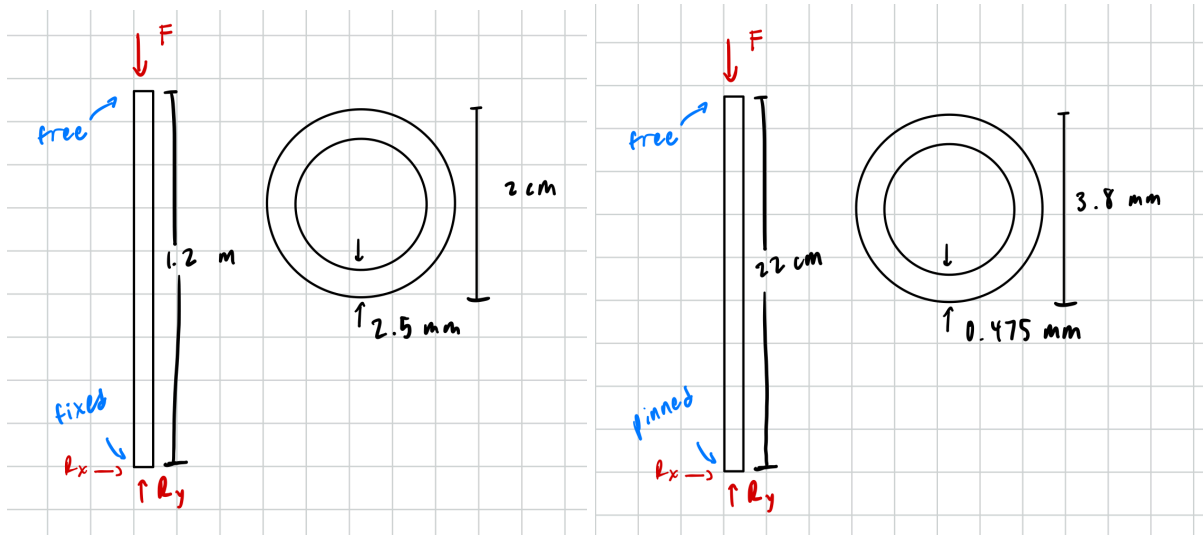


Figure 4: FBD for the life-sized model (left); FBD for the scale-sized model (right)

With only one applied force directly downward at the handle end of the rod, our primary analysis criterion looked at critical buckling force ($P_{cr} = (\pi^2 * E * I) / (K * L)^2$). We chose this criterion because when looking at the yield force for normal stress, the buckling force was significantly smaller, meaning that the buckling force was the limiting factor (Appendix A). This analysis would help us to answer our question of how different ski pole designs differ when loaded (which design could withstand a larger load before buckling). However, for the sake of this project, these analytical results only went toward confirming the FEA and physical testing of the straight pole design since the bent pole was not calculated. In the end, we found that it would take **634.89 N** to buckle the life-sized pole and **2.09 N** to buckle the scaled model (Table 1).

Simulation Results:

For our simulation, we modeled our conditions off of the analytical results section (which was modeled off the physical test for the scaled ski poles). Like the analytical results, we treated both poles like a rod, neglecting the point and handle. This was due to the fact that Fusion returned with an error when running our FEA with our original CAD models (with the handle and tip). Firstly, for the life-sized models, we fixed the bottom and left the top free. We also used Aluminum 6061 to reflect our analytical results. Secondly, for the scale-sized models, we pinned the bottom and left the top free. This ensured that our FEA for the physical test (scale-sized model) reflected our physical testing conditions to the best of our ability given our test setup. We additionally used Polystyrene instead of PLA since Fusion did not have PLA as a material and Polystyrene has a similar Young's Modulus ($E = 3.2 \text{ GPa}$ (Polystyrene) vs $E = 3.3 \text{ GPa}$ (PLA)). For all of the FEA, we used Fusion 360's Structural Buckling Analysis. Here, we applied 1 N of force to the top (free) end of the pole, and looked at Fusion's "Buckling Mode 1" and force

multiplier to determine the minimum force required for buckling. After running our FEA, we found the following results for critical buckling force: straight life-sized = **636.345 N**, bent life-sized = **564.993 N**, straight scale-sized = **1.952 N**, bent scale-sized = **1.165 N** (Table 1).

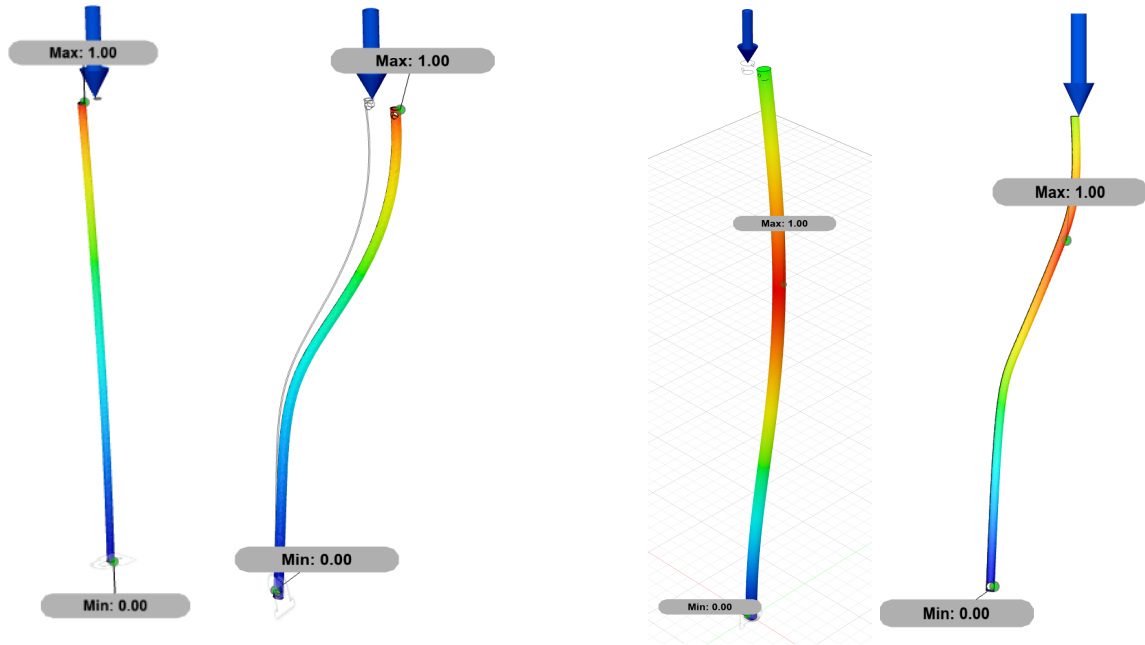


Figure 5: Full-size FEA of straight and bent pole respectively (left, left-center); Scale-size FEA of straight and bent poles respectively (right-center, right)

Prototype Testing:

The first step in our testing was to fabricate our ski poles. Although we originally wanted to do a more complex fabrication with different kinds of metal, we ended up settling on a 3D print using PLA because of our relatively short timeline and likely our need to iterate on the design. Once fabricated, we constrained the poles to a scale on one end with a plastic bushing and a clamp on the other (Figure 6). This setup attempted to model a pinned-free system, where the bushing acted as a pin-like constraint and the top of the clamp acted as a free constraint. We then measured the applied force at the top of the clamp with the scale on the bottom of the pole to measure how much force we were putting through the pole at any time. Videoing tightening the clamp, we visually decided when the poles seemed to begin buckling and then recorded the value on the scale as the corresponding buckling value, converting the grams to Newtons. Throughout this testing process, there were many limitations that could have impacted the accuracy of our results. Firstly, due to the nature of the clamp, we were unable to steadily increase the pressure, rather increasing the pressure in small increments as we squeezed to tighten it. Additionally, the scale itself would update in increments every second, so we were potentially not getting the most accurate results as to what the exact force was on the pole at any given time. Furthermore, given that we visually decided when the poles first started to buckle, this could have resulted in a misreading of the critical bending force. Finally, it was difficult for us to fully model a pinned or

fixed constraint given what we had. This, however, was addressed in the analytical results section. As a result of this test, we found that the straight pole buckled at **1.87 N** and the bent pole at **1.28 N** (Table 1).



Figure 6: Testing set up (left); Bent pole test at critical buckling force (middle); Straight pole test at critical buckling force (right)

Discussion:

Table 1: All Test Results

Pole Design	Analytical Results	FEA Results	Physical Test Results
Life-sized Straight	634.89 N	636.345 N	-
Life-sized Bent	-	564.993 N	-
Scale-sized Straight	2.09 N	1.952 N	1.87 N
Scale-sized Bent	-	1.165 N	1.28 N

From our results, we found that ski pole design significantly influences force propagation when loaded. Specifically, our analysis demonstrates that the straight ski pole design had a higher critical buckling force than the bent ski pole design. This supports our initial intuition that the ski pole design would impact the force propagation through it. However, the degree to which the bent ski pole design decreased the critical buckling load was greater than we initially thought.

Firstly, comparing the analytical and FEA results for the life-sized straight pole, these results were highly consistent, with only a 0.22% difference between the two values (634.89 N vs. 636.345 N). These results are expected as the FEA and analytical processes all had ideal conditions (fixed-free constraint, uniform rod, etc.). However, since we were unable to physically test the life-sized model due to fabrication constraints, we could not validate these results.

Secondly, for the scaled-sized straight pole, the FEA buckling force (1.952 N) closely matched the analytical result (2.09 N) with only a 6.82% difference. This is mainly attributed to the FEA analysis using Polystyrene ($E = 3.2$ GPa) versus the analytical result using PLA ($E = 3.3$ GPa). Additionally, the physical test result (1.87 N) was slightly lower than both these values. This could have been attributed to imperfections/differences in the fabrication (3D printer's layering of the PLA) as well as the other limitations mentioned previously in our testing setup (boundary conditions, visual buckling criterion, test setup, etc).

Finally, for the bent poles, although we could not calculate the analytical results due to the pole's geometry, both FEA and physical testing results demonstrated that the bent pole design had a significantly lower critical buckling force than the straight pole design: 564.993 N vs. 636.345 N for life-sized FEA; 1.165 N vs. 1.952 N for scale-sized FEA; 1.28 N vs 1.87 N for scale-sized physical test. Unexpectedly though, when looking at the physical test results, the bent pole had a higher critical buckling force (1.28 N) than the predicted FEA critical buckling force (1.165 N). This may have been again attributed to limitations in our testing setup, 3D printing's layering implications, and the FEA's use of Polystyrene versus PLA.

Overall, by comparing these three testing methods, we were able to get a generalized idea of how the straight ski pole compared to the bent ski pole when looking at critical bending force. Using three different methods of analysis affirmed our findings and allowed to us see where certain methods were limited in their approach (assumptions, setups, unaccounted variables, etc.). Additionally, using three different methods allowed us to test for the critical buckling force even when certain aspects limited the available tests (e.g. physical testing for the life-sized model and analytical results for both bent poles). Lastly, in an attempt to get more analogous results, we could improve the accuracy of our testing results by modeling the actual boundary conditions, using a homogenous material like Aluminium 6061, or using higher-resolution/larger-scaled 3D printed models.

Final Product:

Due to our fabrication limitations, our final prototype is our scaled-down 3D printed ski-poles, with both the flange and pointed tip removed for physical testing (Figure 7).

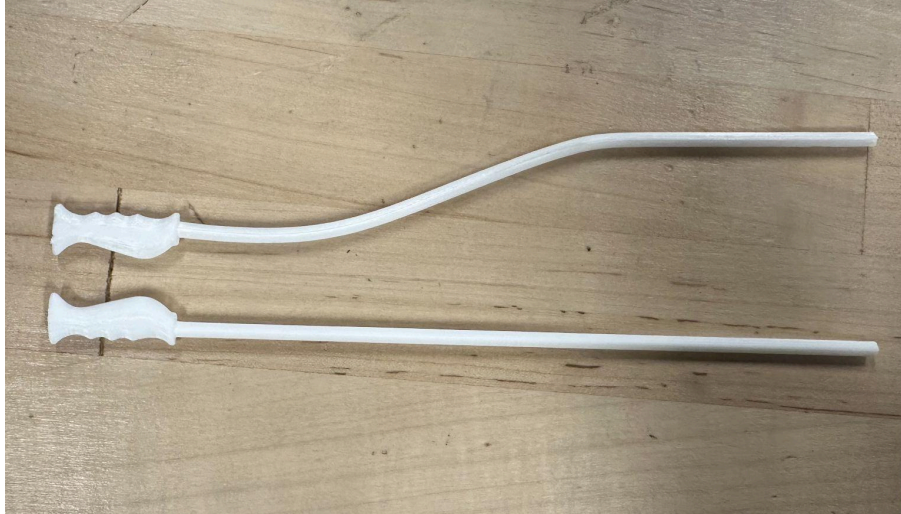


Figure 7: Final product

Conclusion:

Overall, through this project, we successfully reached our goal of finding out how different ski pole designs compared when looking at the critical buckling force. Specifically, the straight ski pole design had a higher critical buckling force than the bent ski pole design. Upon further research, these findings make sense as the curved design of the ski poles is intended for more practicality in slalom skiing where the poles need to wrap around the skier's body for more efficiency and aerodynamics.

One of the biggest things we learned as a team through this project was how difficult it is to take something from hand calculations to computer testing to real-life testing while minimizing the different errors and assumptions that take place. Because of this, it was quite difficult to ensure that the different results we were finding were valid and useable. We had to consider many details we never thought about at the beginning when we planned everything out. We were able to iterate through our different tests to get numbers that were not very far off from one another, but if we had kept our original results for everything, the data would have been quite different. Through these iterations for every type of test, we internalized the concept of buckling. Specifically, we learned how much of a factor length and the K-value (boundary conditions) impact the critical buckling force.

Looking back, we see that having a clearer testing plan could've cut down on trial and error. We also learned how important it is to take accurate measurements and understand how many different factors can affect results. For this assignment specifically, if we wanted to continue our testing, the next step would be building a more accurate life-size model out of more robust materials. Once we had this done, we could go back and calculate both our computer and hand calculations to hopefully get more accurate numbers. We would then need to test this new interaction with a more robust system that could more accurately test the forces applied.

Peer Review:

Brayden: Overall, I think that my group did a great job on this project. I think we all initially were a little in over our heads, meaning that we didn't really know what to expect or how to approach this project, but we all worked together and were willing to iterate on our different parts to continue to finetune our results.

Mari: Our team worked very well together. We all had our tasks but helped each other by sharing ideas. I focused on creating and improving the CAD models and often found myself communicating with the team member doing FEA analysis who gave me feedback on how to adjust the model to make it work better.

Kaimana: Overall, I think our group worked very cohesively. Along the way there were multiple barriers we had to overcome (failed testing, 3D printing errors, calculation errors); however, we got really good at communicating what needed to be changed in each part of the testing and worked cohesively to make sure each part came together. Specifically, I was responsible for the physical prototype and testing, and I found myself communicating with the CAD lead to fix different scaling issues and the FEA lead to see how closely we could replicate the testing boundary conditions on Fusion.

Appendix

Appendix A - Hand Calculations for Analytical Results



Assume : r is constant
 - Pinned-free ends
 - $K = 1.5$
 - treat ski pole as a rod.

Given - PLA ($E = 3.3 \text{ GPa}$)
 - Outer diameter = 3.8 mm
 - Wall thickness = 0.475 mm
 - $L = 220 \text{ mm}$

Find Buckling force (P_{cr})

Solution $P_{cr} = \frac{\pi^2 EI}{(KL)^2}$

$$I = \frac{\pi}{4} (r_o^4 - r_i^4) \Rightarrow r_o = 1.9 \text{ mm} = 0.0019 \text{ m}$$

$$r_i = 1.425 \text{ mm} = 0.001425 \text{ m}$$

$$I = 7.0 \times 10^{-12} \text{ m}^4$$

$$E = 3.3 \text{ GPa} = 3.3 \times 10^9 \text{ N/m}^2$$

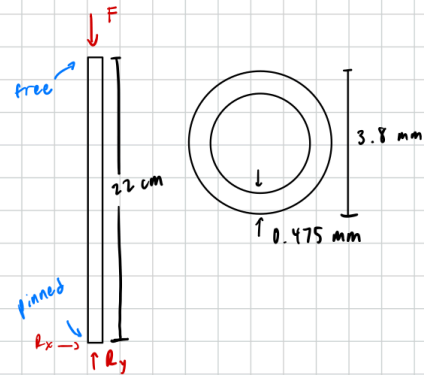
$$P_{cr} = \frac{\pi^2 (3.3 \times 10^9) (7.0 \times 10^{-12})}{(1.5 \cdot 0.22)^2} = 2.09 \text{ N} \text{ (critical buckling force)}$$

$$\sigma = \frac{F}{A} \Rightarrow \sigma_{failure} = 50 \text{ MPa} = \frac{F}{\pi(r_o^2 - r_i^2)}$$

$$F = (50 \times 10^6 \text{ N/m}^2) \pi (0.0019^2 - 0.001425^2)$$

$$F_{failure} = 248 \text{ N}$$

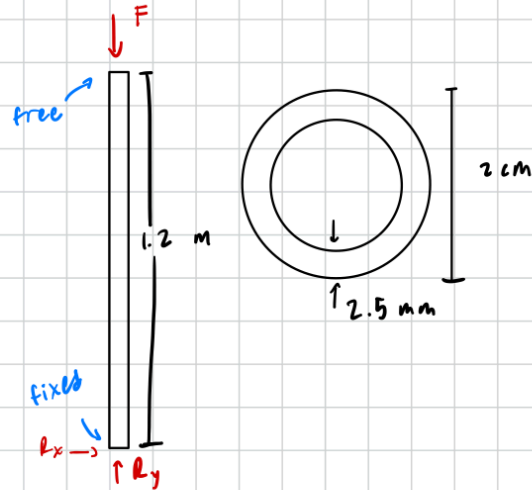
↑ demonstrates that the pole will buckle before yielding



Assume - r is constant
 - fixed-free ends
 - treat ski pole as a rod.

Given - Aluminum 6061 ($E = 69 \text{ GPa}$)
 - Outer diameter = 2 cm
 - Wall thickness = 2.5 mm
 - $L = 1.2$
 - $k = 2$ (fixed-free)

Find Buckling force



$$P_{cr} = \frac{\pi^2 EI}{(kL)^2}$$

$$I = \frac{\pi}{4} (r_o^4 - r_i^4)$$

$$I = \frac{\pi}{4} (0.01^4 - 0.0075^4)$$

$$I = 5.37 \times 10^{-9} \text{ m}^4$$

$$E = 69 \text{ GPa} = 69 \times 10^9 \text{ N/m}^2$$

$$P_{cr} = \frac{\pi^2 (69 \times 10^9) (5.37 \times 10^{-9})}{(2 \times 1.2)^2} = 634.89 \text{ N} \text{ (critical buckling force)}$$

$$\sigma = \frac{F}{A} \Rightarrow \sigma_{\text{Failure}} = 255 \text{ MPa} = \frac{F}{\pi(r_o^2 - r_i^2)}$$

$$F = (255 \times 10^6 \text{ N/m}^2) \pi ((0.01 \text{ m})^2 - (0.0075 \text{ m})^2)$$

$$F = 35.0 \text{ kN}$$

↑ demonstrate pole will buckle before yielding